USACE Savannah District: Mathematical & Regulatory Framework

Quantitative Credit Calculations, Developer Debit Formulas, & RIBITS Ledger Mechanics

This framework establishes the exact mathematical models, credit/debit worksheets, and administrative processes utilized by the **U.S. Army Corps of Engineers (USACE) Savannah District** to govern stream compensatory mitigation in Georgia. It serves as Hunter's definitive guide to calculating credit yields and understanding how developers are mathematically assessed for their environmental impacts.

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1. How the U.S. Army Corps of Engineers (USACE) Operates

The USACE is organized geographically into divisions and districts. Georgia falls under the **South Atlantic Division (SAD)**, and the **Savannah District** holds regulatory jurisdiction over all wetlands and streams in the state.
The Regulatory Branch of the Savannah District operates under the **2008 Compensatory Mitigation Rule** (33 CFR Parts 325 and 332). The District is co-chaired by the **Interagency Review Team (IRT)**, which includes:

- U.S. Army Corps of Engineers (USACE) — *Savannah District (Chair)*
- U.S. Environmental Protection Agency (EPA) — *Region 4*
- U.S. Fish and Wildlife Service (USFWS)
- National Marine Fisheries Service (NMFS)
- Georgia Department of Natural Resources (DNR) — *Environmental Protection Division (EPD)*

All commercial mitigation banks must be approved by the IRT through a consensus process, whereas Permittee-Responsible Mitigation (PRM) projects (such as Anderson Creek) are approved directly by the USACE Project Manager in coordination with EPD stream buffer variance regulators.

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2. The Developer's Debit Formula (Calculating Required Credits)

When a developer (such as **Fuqua Development** or **Sefried Development**) impacts a stream, the USACE Savannah District assesses the impact using the **Adverse Impact Worksheet**. This worksheet calculates the total number of **debits** (credits the developer must purchase from a bank to achieve compliance).

The Mathematical Model for Adverse Impact (Debits)

$$\text{Total Debits Required (D)} = \text{Stream Length Impacted (LF)} \times \sum (\text{Impact Factors}) \times R$$

Where:

- ***$\text{Stream Length Impacted (LF)}$***: The linear footage of stream channel physically disturbed, piped, filled, or culverted.

- ***$\sum (\text{Impact Factors})$***: The sum of values assigned to four distinct qualitative and quantitative parameters:

1. **Stream Type Factor (*F_{type}*)**: Measures the flow regime of the impacted stream.
- Perennial (flowing year-round): **0.9**
- Intermittent (flowing seasonally): **0.4**
- Ephemeral (flowing only after rain): **0.1**
2. **Impact Type Factor (*F_{impact}*)**: Measures the severity of the structural impact.
- Piping/Culverting (complete stream loss): **2.0**
- Channelization/Relocation (concrete or geogrid lining): **1.5**
- Clearing/Grubbing (vegetation removal only): **0.5**
3. **Duration of Impact Factor (*F_{duration}*)**:
- Permanent (structure remains in perpetuity): **1.5**
- Temporary (< 1 year for utility crossing): **0.2**
4. **Watershed Priority Factor (*F_{priority}*)**: High-priority basins identified in the district's River Basin Management Plans.
- Primary Priority Watershed (e.g., active trout streams): **0.4**
- Secondary Priority Watershed: **0.1**
- ***R* (Regional Multiplier)**: Savannah District regional scaling factor (typically set to **1.0**).

■ Practical Example: Developer Debit Calculation

Consider a project proposed by **Fuqua Development** that requires piping **1,200 linear feet** of a perennial trout stream in a high-priority watershed (e.g., Coosa River basin) to build a retail shopping center:

1. **Parameters**:
- Stream Length = $1,200\text{ LF}$
- Stream Type (F_{type}) = Perennial $\rightarrow$ **0.9**
- Impact Type (F_{impact}) = Piping $\rightarrow$ **2.0**
- Duration (F_{duration}) = Permanent $\rightarrow$ **1.5**
- Priority (F_{priority}) = Primary $\rightarrow$ **0.4**
2. **Sum of Factors**:

$$\sum (\text{Factors}) = 0.9 + 2.0 + 1.5 + 0.4 = 4.8$$

3. **Total Debits Required**:

$$\text{Debits} = 1,200\text{ LF} \times 4.8 = \mathbf{5,760\text{ Credits}}$$

4. **Financial Cost to Developer**:

If stream credits in the basin trade at **$110/credit**, Fuqua Development must purchase $5,760\text{ credits}$ at a total cost of **$633,600** from a mitigation bank to secure their permit.

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3. The Bank's Credit Formula (Calculating Credit Generation)

To determine how many **credits** Hunter’s stream restoration projects generate, the USACE Savannah District applies the **Mitigation Credit Worksheet**. This worksheet calculates the total ecological lift based on instream restructuring and riparian buffer enhancements.

The Mathematical Model for Stream Mitigation Credits (Credits)

$$\text{Credits Generated (C)} = \text{Restored Length (LF)} \times \sum (\text{Mitigation Factors}) \times W$$

Where:

- **$\text{Restored Length (LF)}$**: The linear footage of stream channel physically restored.
- **$\sum (\text{Mitigation Factors})$**: The sum of values assigned to six ecological lifting parameters:

1. **Net Benefit Factor (M_{benefit})**: The scale of geomorphic and biological lift.

- *Priority 1 Restoration* (Rosgen Natural Channel Design, reconnecting stream to historic floodplain footing): **4.0**
- *Priority 2 Restoration* (Creating a new, stable floodplain bench at current elevation): **3.0**
- *Priority 3 / Bioengineering* (Bank grading and soft armoring with live stakes/root wads): **1.5**

2. **Riparian Buffer Factor (M_{buffer})**: Evaluated per bank (total of Left + Right banks).

- 100-foot buffer planted with native woody species: **1.2 per bank** (Total: **2.4**)
- 50-foot buffer planted with native woody species: **0.6 per bank** (Total: **1.2**)

3. **Monitoring Factor ($M_{\text{monitoring}}$)**:

- 7-Year Trout Stream Commitment (including thermal and EPT macroinvertebrate scoring): **0.4**
- 5-Year Standard Commitment: **0.2**

4. **Site Protection Factor ($M_{\text{protection}}$)**:

- Perpetual Restrictive Covenant/Easement held by third-party Land Trust: **1.0**

5. **Stream Flow Factor (M_{flow})**:

- Perennial: **0.4**
- Intermittent: **0.2**

6. **Watershed Priority Factor ($M_{\text{watershed}}$)**:

- Primary Priority Watershed (e.g., North Georgia trout waters): **0.4**
- ***W* (Watershed Multiplier)**: Adjusts credit yield based on structural location (typically set to **1.0**).

■ Practical Example: Anderson Creek (ACR) Credit Yield

Using Hunter's geomorphic design for the **30-acre Anderson Creek lot** across **1,500 linear feet** of degraded perennial stream:

1. **Parameters**:

- Stream Length = $1,500\text{ LF}$
- Net Benefit (M_{benefit}) = Priority 1 Geomorphic Restoration $\rightarrow$ **4.0**
- Riparian Buffer (M_{buffer}) = 100-foot buffer on both banks $\rightarrow 1.2 \times 2 = \mathbf{2.4}$
- Monitoring ($M_{\text{monitoring}}$) = 7-Year Trout SOP $\rightarrow$ **0.4**
- Site Protection ($M_{\text{protection}}$) = Perpetual Conservation Easement $\rightarrow$ **1.0**

- Stream Flow (M_{flow}) = Perennial $\rightarrow$ **0.4**
- Watershed Priority ($M_{\text{watershed}}$) = Primary Trout Watershed $\rightarrow$ **0.4**

2. **Sum of Mitigation Factors**:

$$\sum (\text{Factors}) = 4.0 + 2.4 + 0.4 + 1.0 + 0.4 + 0.4 = \mathbf{8.6}$$

3. **Total Credits Generated**:

$$\text{Credits} = 1{,}500\text{ LF} \times 8.6 = \mathbf{12{,}900\text{ Credits}}$$

4. **Gross Asset Value**:

At a premium coldwater trout basin credit price of **$180/credit** (due to extreme scarcity in North Georgia), the total credit asset generated from the Anderson Creek lot is **$2,322,000**!

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4. RIBITS and Ledger Mechanics

All stream and wetland mitigation banks are tracked in a public federal database called **RIBITS** (Regulatory In-lieu fee and Bank Information Tracking System), maintained by the USACE.

RIBITS Ledger Operations

- **Credit Ledger**: Every approved bank has a credit ledger in RIBITS. Credits are added to the ledger upon reaching specific regulatory milestones (credit releases) and are deducted when sold to developers.
- **Credit Releases**: Credits are not released all at once. The Savannah District operates under a standard **Credit Release Schedule**:

1. **Initial Release (15%)**: Released immediately upon signing the Mitigation Banking Instrument (MBI), recording the perpetual Conservation Easement, and funding the Financial Assurances. (At Anderson Creek, this equals **1,935 credits** or **$348,300** in immediate tradable value).
2. **Construction Release (25%)**: Released upon completion of all instream geomorphic earthwork and USACE approval of the as-built survey.
3. **Annual Success Releases (60%)**: Released in equal increments of **10%** over the subsequent 6 years, contingent on satisfying the geomorphic stability, vegetation survival, and water temperature success criteria.

Financial Assurances

To secure credit releases, the Bank Sponsor must establish a **Financial Assurance Account** (typically in the form of a **Performance Bond** or **Letter of Credit** held in escrow).

- **Purpose**: Protects the USACE in case the bank sponsor goes bankrupt or fails to maintain the stream channel during the 7-year monitoring phase.
- **Funding**: The bond is typically funded at a rate of **$15 to $25 per linear foot** of restored stream. For Anderson Creek, this represents a capital bond of **\sim 30,000 to $50,000**, which is fully returned upon completing the 7-year monitoring period successfully.